Data Mining in Python

Today's goal • 3/3 items

Module 1
Basic Concepts of Data Mining

Welcome to the Course

- Welcome to Data Mining in Python
  Video • 2 min
- MADSwPY Certificate Roadmap
  Reading • 5 min
- Course Syllabus
  Reading • 10 min
- Meet Your Fellow Learners
  Discussion Prompt • 15 min
- Help Us Learn About You
  Reading • 10 min

Lesson 1: Introduction to Data Mining and Its Basic Functionalities

- What is Data Mining
  Video • 14 min
- Introduction to the Basic Functionalities of Data Mining
  Reading • 10 min
- Data Mining Functionalities (Part 1)
  Video • 6 min
- Data Mining Functionalities (Part 2)
  Video • 7 min
- Data Mining Functionalities (Part 3)
  Video • 4 min
- Knowledge Check: Basic Functionalities of Data Mining
  Practice Assignment • 15 min

Lesson 2: Representations of Real World Data

- Introduction to Basic Data Representations
  Reading • 10 min
- Representing Itemsets, Vectors, and Matrices
  Video • 6 min
- Representing Sequences
  Video • 4 min
- Knowledge Check: Basic Data Representations (Part 1)
  Practice Assignment • 15 min
- Representing Time-Series and Spatial/Temporal Data
  Video • 8 min
- Representing Graph Data
  Video • 5 min
- Representing Stream Data
  Video • 4 min
- Case Study: Representations of Real-World Text Data
  Reading • 10 min
- Knowledge Check: Basic Data Representations (Part 2)
  Practice Assignment • 15 min

Lesson 3: Patterns and Similarities

- Introduction to Patterns and Similarities
  Reading • 10 min
- Data Mining Based on Patterns
  Video • 5 min
- Data Mining Based on Similarities
  Video • 6 min

Module 1 Wrap Up

- Module 1 Quiz: Introduction to Data Mining
  Graded Assignment • 20 min
- Introduction to Module 1 Programming Assignment: Visualizing Different Data
  Reading • 10 min
- Module 1 Programming Assignment: Warming Up
  Programming Assignment • 3h
- Dialogue Reflection
  Dialogue • 15 min
- Module 1 Optional Readings & Resources
  Reading • 10 min

Module 2
Mining Itemset Data

Introduction to the Basic Functionalities of Data Mining

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The Four Dimensions of Data Mining

You may have seen different views of data mining from different practices. When you have a data mining task, you can usually describe the task using **four** dimensions:

- **First Dimension: Data to Be Mined**, which is actually the ***input*** of the data mining task.
- **Second Dimension: Knowledge to Be Discovered**, which is what type of patterns you are trying to discover from the data, thus, the ***output*** of your data mining task.
- **Third Dimension: Techniques Utilized** to **connect the *input*** to the ***output***.
- **Fourth Dimension: Applications Adopted** in which people adapt this data mining process, thus, **where to use this data.**

Data to Be Mined

To think about what kind of data can data mining process take as input. We can characterize real-world data based on either their formal types or their representations.

- **Type/representation:** We can divide data into itemsets, vectors or matrices, sequences, time series, spatiotemporal data, data streams, graphs, and networks.
- **Genre/application**: Taking a different perspective, we could also divide real-world data based on the genre or application domains. If you take this view, you can categorize data as transactional data, text data, web data, multimedia data, social and information networks, biological data, or user behaviors.

This application-driven categorization is also very useful for real-world tasks.

Knowledge to Be Discovered (also known as 'data mining functionalities')

What functionality do you want to provide through a data mining process?

These functionalities could include lower-level outputs such as patterns from the data. similarities of data, or associations of data. The functionalities also include higher-level outputs that could be directly used to support decisions, for example, classifications, clustering, trends, predictions, and outliers on an even higher level.

Based on the goal of data mining, whether you are trying to describe or characterize that data set or whether you are trying to make predictions, a data mining process can also be divided into ***descriptive*** or ***predictive***.

Techniques Utilized

Once you define the input and output over a data mining process, you can talk about the techniques that connect the input to the output. Particular techniques used in the data mining process include data cubing, machine learning, statistics, pattern recognition, user modeling, information visualization, and data-intensive computing.

Applications Adopted

Once you can generate the output from the input, the data mining process can be applied in many different domains. Here are a few examples:

- In retail, data mining is used to do intelligent advertisements—to segment their market.
- In telecommunications, data mining is used to detect spam calls, which, unfortunately, we are not doing very well nowadays.
- In banking, data mining is used to make decisions on unknowns such as the applicants' credit scores which can be estimated through a data mining process (a decision on whether to approve a loan or not can be made)
- In social media, data mining is used on Facebook and Twitter. They use data mining to recommend friends, and they rank the feeds on those social networks.
- In science, data mining is used to drive scientific discoveries. This is very much related to Jim Gray's, The Fourth Paradigm of Science.
- Google uses data mining to answer complicated questions to provide personalized search results to the users.
- Wall Street also uses data mining to build portfolios of stocks.
- Data mining is also applied to text using techniques such as Natural Language Processing to make sense of text data.
- Medical practitioners also use data mining to make sense of patient data. Data mining has been one of the technical foundations of health informatics nowadays.

These are just some examples of application domains of data mining, as well as many others.

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